

ME563 HW 3

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Problem 1

Problem statement. The components of the mass flow vector $\rho \vec{u}$ are $\rho u_x = 4x^2y$, $\rho u_y = xyz$, and $\rho u_z = yz^2$.

(a)

Problem statement. Compute the net mass outflow through the closed surface formed by the planes, $x = 0, x = 1, y = 0, y = 1, z = 0, z = 1$.

$\rho \mathbf{u} = \langle 4x^2y, xyz, yz^2 \rangle, \Phi = \oint_S \rho \mathbf{u} \cdot \hat{\mathbf{n}} dA$:

$$x = 0, \hat{\mathbf{n}} = -\hat{\mathbf{e}}_x : \Phi = \int_0^1 \int_0^1 -4x^2y|_{x=0} dz dy = 0$$

$$x = 1, \hat{\mathbf{n}} = +\hat{\mathbf{e}}_x : \Phi = \int_0^1 \int_0^1 4y dz dy = 4 \int_0^1 y dy = 2$$

$$y = 0, \hat{\mathbf{n}} = -\hat{\mathbf{e}}_y : \Phi = \int_0^1 \int_0^1 -xyz|_{y=0} dz dx = 0$$

$$y = 1, \hat{\mathbf{n}} = +\hat{\mathbf{e}}_y : \Phi = \int_0^1 \int_0^1 xz dz dx = \frac{1}{2} \int_0^1 x dx = \frac{1}{4}$$

$$z = 0, \hat{\mathbf{n}} = -\hat{\mathbf{e}}_z : \Phi = \int_0^1 \int_0^1 -yz^2|_{z=0} dy dx = 0$$

$$z = 1, \hat{\mathbf{n}} = +\hat{\mathbf{e}}_z : \Phi = \int_0^1 \int_0^1 y dy dx = \frac{1}{2} \int_0^1 dx = \frac{1}{2}$$

$$\boxed{\oint_S \rho \mathbf{u} \cdot \hat{\mathbf{n}} dA = 0 + 2 + 0 + \frac{1}{4} + 0 + \frac{1}{2} = \frac{11}{4}}$$

(b)

Problem statement. Compute $\vec{\nabla} \cdot (\rho \vec{u})$ and integrate over the volume bounded by the surface defined in part a).

$$\nabla \cdot (\rho \mathbf{u}) = \frac{\partial}{\partial x}(4x^2y) + \frac{\partial}{\partial y}(xyz) + \frac{\partial}{\partial z}(yz^2) = \boxed{8xy + xz + 2yz}$$

$$\begin{aligned}
\int_0^1 \int_0^1 \int_0^1 (8xy + xz + 2yz) dz dy dx &= \int_0^1 \int_0^1 \left(8xyz + \frac{1}{2}xz^2 + yz^2 \right) \Big|_{z=0}^{z=1} dy dx \\
&= \int_0^1 \int_0^1 \left(8xy + \frac{1}{2}x + y \right) dy dx \\
&= \int_0^1 \left(4xy^2 + \frac{1}{2}xy + \frac{1}{2}y^2 \right) \Big|_{y=0}^{y=1} dx = \int_0^1 \left(4x + \frac{1}{2}x + \frac{1}{2} \right) dx \\
&= \left(2x^2 + \frac{1}{4}x^2 + \frac{1}{2}x \right) \Big|_{x=0}^{x=1}
\end{aligned}$$

$$\boxed{\int_V \nabla \cdot (\rho \mathbf{u}) dV = \frac{11}{4}}$$

(c)

Problem statement. Explain why the results in a) and b) should, or should not, be equal.

They should be equal. $\rho \mathbf{u}$ is continuously differentiable on the cube V , so the divergence theorem applies:

$$\int_V \nabla \cdot (\rho \mathbf{u}) dV = \oint_S \rho \mathbf{u} \cdot \hat{\mathbf{n}} dA = \frac{11}{4}.$$

Problem 2

Problem statement. For the following flows find the missing velocity component needed for the flow to satisfy the incompressible continuity equation

Incompressible continuity: $\nabla \cdot \mathbf{u} = \frac{\partial u_1}{\partial x_1} + \frac{\partial u_2}{\partial x_2} + \frac{\partial u_3}{\partial x_3} = 0$. In each case f is an arbitrary function of the two coordinates not differentiated against.

(a)

Problem statement. $u_1 = x_1^2 + x_2^2 + a^2$, $v_2 = -x_1x_2 - x_2x_3 - x_1x_3$, $u_3 = ?$

(v_2 taken as u_2 . I think this was a typo in the problem statement.)

$$\begin{aligned}
\frac{\partial}{\partial x_1} (x_1^2 + x_2^2 + a^2) + \frac{\partial}{\partial x_2} (-x_1x_2 - x_2x_3 - x_1x_3) + \frac{\partial u_3}{\partial x_3} &= 0 \\
\Rightarrow 2x_1 - x_1 - x_3 + \frac{\partial u_3}{\partial x_3} &= 0 \Rightarrow \frac{\partial u_3}{\partial x_3} = x_3 - x_1
\end{aligned}$$

$$\int du_3 = \int (x_3 - x_1) dx_3 \Rightarrow \boxed{u_3 = \frac{1}{2}x_3^2 - x_1x_3 + f(x_1, x_2)}$$

(b)

Problem statement. $u_1 = \ln(x_2^2 + x_3^2)$, $u_2 = \sin(x_1^2 + x_3^2)$, $u_3 = ?$

$$\frac{\partial}{\partial x_1} [\ln(x_2^2 + x_3^2)] + \frac{\partial}{\partial x_2} [\sin(x_1^2 + x_3^2)] + \frac{\partial u_3}{\partial x_3} = 0 \Rightarrow 0 + 0 + \frac{\partial u_3}{\partial x_3} = 0 \Rightarrow \boxed{u_3 = f(x_1, x_2)}$$

(c)

Problem statement. $u_1 = ?$, $u_2 = \frac{x_2}{(x_1^2 + x_2^2 + x_3^2)^{3/2}}$, $u_3 = \frac{x_3}{(x_1^2 + x_2^2 + x_3^2)^{3/2}}$

Let $r = (x_1^2 + x_2^2 + x_3^2)^{1/2}$, so $\partial r / \partial x_i = x_i / r$ and

$$\frac{\partial}{\partial x_i} \left(\frac{x_i}{r^3} \right) = \frac{1}{r^3} - \frac{3x_i}{r^4} \frac{x_i}{r} = \frac{r^2 - 3x_i^2}{r^5} \quad (\text{no sum}).$$

Continuity:

$$\frac{\partial u_1}{\partial x_1} + \frac{r^2 - 3x_2^2}{r^5} + \frac{r^2 - 3x_3^2}{r^5} = 0 \Rightarrow \frac{\partial u_1}{\partial x_1} = \frac{3(x_2^2 + x_3^2) - 2r^2}{r^5} = \frac{3(r^2 - x_1^2) - 2r^2}{r^5} = \frac{r^2 - 3x_1^2}{r^5}$$

which is $\partial(x_1/r^3)/\partial x_1$. Hence

$$\int du_1 = \int \frac{r^2 - 3x_1^2}{r^5} dx_1 \Rightarrow \boxed{u_1 = \frac{x_1}{(x_1^2 + x_2^2 + x_3^2)^{3/2}} + f(x_2, x_3)}$$

Problem 3

Problem statement. Consider the velocity field $\mathbf{u} = (\alpha x/t, 0, 0)$ (in Cartesian), where α is a constant. The density of the fluid is spatially uniform, but time-dependent, i.e., ρ is a function of t .

(a)

Problem statement. Determine the density field $\rho(t)$, given $\rho = \rho_0$ at $t = t_0$.

$\mathbf{u} = \begin{bmatrix} \alpha x/t & 0 & 0 \end{bmatrix}^\top$, $\alpha \in \mathbb{R}$, $\rho = \rho(t)$. Since $\nabla \rho = \mathbf{0}$, continuity reduces to the form discussed in lecture

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{u}) = 0 \Rightarrow \frac{\partial \rho}{\partial t} + \rho \left[\frac{\partial}{\partial x} \left(\frac{\alpha x}{t} \right) + 0 + 0 \right] = 0 \Rightarrow \frac{d\rho}{dt} = -\rho \frac{\alpha}{t}$$

$$\int_{\rho_0}^{\rho} \frac{d\rho'}{\rho'} = -\alpha \int_{t_0}^t \frac{dt'}{t'} \Rightarrow \ln \frac{\rho}{\rho_0} = -\alpha \ln \frac{t}{t_0} = \ln \left(\frac{t_0}{t} \right)^\alpha \Rightarrow \boxed{\rho(t) = \rho_0 \left(\frac{t_0}{t} \right)^\alpha}$$

(b)

Problem statement. What are the temporal ($\partial \mathbf{u} / \partial t$), advective ($\mathbf{u} \cdot (\nabla \mathbf{u})$), and total ($D\mathbf{u} / Dt$) accelerations for this flow field? What does $\alpha = 1$ imply?

Temporal:

$$\frac{\partial \mathbf{u}}{\partial t} = \frac{\partial}{\partial t} \left(\frac{\alpha x}{t} \right) \hat{\mathbf{e}}_x \Rightarrow \boxed{\frac{\partial \mathbf{u}}{\partial t} = -\frac{\alpha x}{t^2} \hat{\mathbf{e}}_x}$$

Advective:

$$\mathbf{u} \cdot \nabla \mathbf{u} = u_x \frac{\partial u_x}{\partial x} \hat{\mathbf{e}}_x = \frac{\alpha x}{t} \cdot \frac{\alpha}{t} \hat{\mathbf{e}}_x \Rightarrow \boxed{\mathbf{u} \cdot \nabla \mathbf{u} = \frac{\alpha^2 x}{t^2} \hat{\mathbf{e}}_x}$$

Total:

$$\frac{D\mathbf{u}}{Dt} = \frac{\partial \mathbf{u}}{\partial t} + \mathbf{u} \cdot \nabla \mathbf{u} \Rightarrow \boxed{\frac{D\mathbf{u}}{Dt} = \frac{\alpha(\alpha-1)x}{t^2} \hat{\mathbf{e}}_x}$$

$\alpha = 1$:

$$\frac{\partial \mathbf{u}}{\partial t} = -\frac{x}{t^2} \hat{\mathbf{e}}_x, \quad \mathbf{u} \cdot \nabla \mathbf{u} = +\frac{x}{t^2} \hat{\mathbf{e}}_x, \quad \boxed{\frac{D\mathbf{u}}{Dt} = \mathbf{0}}$$

The local and advective accelerations cancel each other out; hence, the flow field is unsteady at every point, but every fluid particle moves at **constant velocity**.

Problem 4

Problem statement. Consider a two-dimensional flow with velocity components given by

$$\begin{aligned} v_1 &= c x_1 \\ v_2 &= -c x_2 \end{aligned}$$

Find expressions for the vorticity and the strain rate tensor, (i.e. calculate their components)

Velocity gradient tensor:

$$\frac{\partial v_i}{\partial x_j} = \begin{bmatrix} \partial v_1 / \partial x_1 & \partial v_1 / \partial x_2 \\ \partial v_2 / \partial x_1 & \partial v_2 / \partial x_2 \end{bmatrix} = \begin{bmatrix} c & 0 \\ 0 & -c \end{bmatrix}$$

Vorticity, $\boldsymbol{\omega} = \nabla \times \mathbf{v}$, $\omega_i = \epsilon_{ijk} \partial_j v_k$. For planar flow ($v_3 = 0$, $\partial / \partial x_3 = 0$), $\omega_1 = \omega_2 = 0$ and

$$\omega_3 = \frac{\partial v_2}{\partial x_1} - \frac{\partial v_1}{\partial x_2} = 0 - 0 \Rightarrow \boxed{\boldsymbol{\omega} = (0, 0, 0)}$$

The flow is irrotational ($R_{ij} = 0$).

Strain rate tensor, $S_{ij} = \frac{1}{2} \left(\frac{\partial v_i}{\partial x_j} + \frac{\partial v_j}{\partial x_i} \right)$:

$$S_{11} = \frac{\partial v_1}{\partial x_1} = c, \quad S_{22} = \frac{\partial v_2}{\partial x_2} = -c, \quad S_{12} = S_{21} = \frac{1}{2} \left(\frac{\partial v_1}{\partial x_2} + \frac{\partial v_2}{\partial x_1} \right) = 0$$

$$\boxed{S_{ij} = \begin{bmatrix} c & 0 \\ 0 & -c \end{bmatrix}}$$

$S_{ii} = c - c = 0$, so I think the flow is simply irrotational straining.

Problem 5

Problem statement. Consider a plane Couette flow of a viscous fluid confined between two flat plates a distance b apart. At steady state the velocity distribution is $u = Uy/b$ and $v = w = 0$, where the upper plate at $y = b$ is moving parallel to itself at speed U , and the lower plate is held stationary. Find the rates of linear strain, the rate of shear strain, and vorticity in the flow.

The only nonzero velocity gradient is $\partial u / \partial y = U/b$:

$$\frac{\partial u_i}{\partial x_j} = \begin{bmatrix} 0 & U/b & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

Rates of linear strain:

$$S_{xx} = \frac{\partial u}{\partial x} = 0, \quad S_{yy} = \frac{\partial v}{\partial y} = 0, \quad S_{zz} = \frac{\partial w}{\partial z} = 0$$

Rate of shear strain:

$$\boxed{\frac{\partial u}{\partial y} + \frac{\partial v}{\partial x} = 2S_{xy} = \frac{U}{b}} \Rightarrow S_{xy} = S_{yx} = \frac{U}{2b}, \quad S_{xz} = S_{yz} = 0$$

$$S_{ij} = \begin{bmatrix} 0 & U/(2b) & 0 \\ U/(2b) & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

Vorticity:

$$\boldsymbol{\omega} = \nabla \times \mathbf{u} = \left(\frac{\partial w}{\partial y} - \frac{\partial v}{\partial z} \right) \hat{\mathbf{e}}_x + \left(\frac{\partial u}{\partial z} - \frac{\partial w}{\partial x} \right) \hat{\mathbf{e}}_y + \left(\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} \right) \hat{\mathbf{e}}_z \Rightarrow \boxed{\boldsymbol{\omega} = -\frac{U}{b} \hat{\mathbf{e}}_z}$$